



ENGINEERED SOLUTIONS
FOR A CLEANER, SAFER TOMORROW

RELIABLE SOLUTIONS.
REAL PROGRESS.

DOSING
PRECISION
ADDITIVES
TRANSFER
CONTROL

ACCURATE
FLEXIBLE
CUSTOMIZABLE

PRECISION MICRO GEAR DOSING SKID SYSTEM LNS-DZ-PDG

Very Low Flow Additive & Specialty Chemical Dosing System

PEOPLE
INDUSTRY
CLEANER
TOMORROW



SPECIALTY
CHEMICALS



FUEL
ADDITIVES



FINE
DOSING



INDUSTRIAL
PROCESS



UTILITIES
& OTHERS

SOLUTIONS
FOR A SAFER TOMORROW

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PRODUCT OVERVIEW & APPLICATION SCOPE

1. PRODUCT DESCRIPTION

The **LINESiG LNS-DZ-PDG** is a precision positive-displacement micro gear dosing skid developed for very low-flow, high-accuracy and repeatable chemical or additive injection. The pump uses closely fitted rotating gears to deliver a near-continuous, low-pulsation flow, making the system suitable for applications where stable metering and fine flow control are more important than very high discharge pressure.

The package is intended for clean, lubricating or suitably filtered liquids and can be engineered for specialty chemicals, solvents, hydrocarbon additives, catalysts, dyes, markers and process-conditioning fluids. It can be supplied as a single-train module or as a multi-pump skid with independent metering and control for each injection point.

2. INDUSTRIES

- Refinery and petrochemical additive systems
- Fuel terminals and fuel-additive blending
- Lubricant and specialty oil dosing
- Chemical and specialty-chemical manufacturing
- Pilot plants, laboratories and process-development systems
- Fine chemical, catalyst and solvent feed systems
- Industrial OEM packages requiring accurate low-flow injection

3. TYPICAL MEDIA

Media Group	Typical Examples	Engineering Consideration
Fuel / Hydrocarbon Additives	Dyes, markers, lubricity improvers, performance additives	Cleanliness, viscosity and lubricity must suit gear pump clearances.
Specialty Chemicals	Catalysts, solvents, inhibitors, modifiers	Material compatibility and filtration are critical.
Lube / Oil Additives	Low-flow blending additives, viscosity modifiers, package components	Temperature may be used to control viscosity where required.
Clean Process Liquids	Selected non-abrasive chemicals and low-solids liquids	Not suitable for slurries or abrasive solids unless specifically engineered.

TYPICAL MICRO GEAR DOSING SKID CONFIGURATION

4. SYSTEM ARRANGEMENT

A typical LNS-DZ-PDG package includes a day tank or remote suction connection, suction isolation and filtration, precision micro gear pump(s), motor/servo drive, high-accuracy flow verification, discharge pressure monitoring, fine control valve where required, check/injection valves and a PLC/HMI or remote control interface.

Subsystem	Typical Arrangement
Pump Configuration	1 x 100%, 1W+1S or multiple independent micro gear pump trains
Suction	Short, low-loss suction line with isolation and fine filtration
Drive	VFD, servo motor or precision variable-speed drive
Flow Verification	Coriolis, precision PD meter or suitable low-flow meter
Pressure Monitoring	Local gauge and/or pressure transmitter
Discharge Control	Speed control primary; optional fine control/injection valve
Automation	Local PLC/HMI, batch, ratio or remote DCS/SCADA

5. PUMP TECHNOLOGY

- External or internal precision gear elements depending on selected pump design
- Smooth rotary displacement with significantly lower pulsation than reciprocating metering pumps
- Compact pump head suitable for modular multi-line skids
- High repeatability with stable fluid viscosity and accurate speed control
- Magnetic-drive or mechanically sealed variants available for selected duties

TECHNICAL SPECIFICATIONS

6. GENERAL DESIGN DATA

Parameter	Typical / Available Basis
Service	Precision dosing, additive injection, fine blending and specialty chemical feed
Flow Range	Project specific; typical package duty approximately 0.1 to 250 L/h per pump
Discharge Pressure	Project specific; commonly 2–20 bar, higher by engineered pump selection
Pump Type	Positive-displacement precision micro gear pump
Flow Characteristic	Near-continuous, low-pulsation rotary displacement
Accuracy / Repeatability	High, subject to calibration, fluid properties and selected meter/control system
Turndown	Wide with precision speed control; final ratio pump/drive dependent
Operating Temperature	According to fluid viscosity and wetted material selection
Wetted Materials	SS316/316L, alloy steels or selected specialty materials
Seal Option	Mechanical seal or magnetic-drive/sealless arrangement where available
Skid Frame	Coated carbon steel or stainless steel
Installation	Indoor/outdoor; hazardous-area execution available by project

7. FLOW SELECTION BASIS

Final pump sizing shall consider minimum and maximum dosing rate, differential pressure, viscosity over the full temperature range, density, lubricity, vapor pressure, solids content, required accuracy and turndown. Gear pumps should not be selected solely from nominal displacement because internal slip changes with viscosity and differential pressure.

FILTRATION, VISCOSITY & SUCTION DESIGN

8. FILTRATION REQUIREMENT

Precision gear pumps rely on close internal clearances. Suction-side filtration is therefore an important part of the package. Filter rating shall be selected from the pump manufacturer's particle-size tolerance, fluid cleanliness and allowable pressure drop. Duplex filtration can be supplied where uninterrupted service is required.

9. VISCOSITY MANAGEMENT

- Flow performance is influenced by fluid viscosity and temperature.
- Heating or heat tracing may be incorporated for viscous additives where required.
- Very low-viscosity fluids can increase internal slip and reduce volumetric efficiency.
- Very high viscosity can increase suction losses and drive torque.
- Final speed range and pump displacement should be verified at minimum and maximum viscosity.

10. SUCTION DESIGN

A flooded or positive suction condition is preferred. Suction piping should be short and adequately sized, with minimum fittings and clean filtration. Pump inlet pressure must remain safely above the fluid vapor pressure to avoid cavitation and loss of metering stability.

METERING & FLOW CONTROL

11. CONTROL PHILOSOPHY

The primary control method for LNS-DZ-PDG is precision speed control. A flowmeter may provide closed-loop correction where higher accuracy is required. The dosing setpoint may be fixed locally, derived as a ratio of the main process flow, or issued by a batch/PLC/DCS system.

Control Function	Typical Implementation
Pump Capacity Control	VFD or servo speed control
Flow Measurement	Coriolis or high-accuracy low-flow positive-displacement meter
Ratio Dosing	Setpoint proportional to main process flow
Batch Control	Preset quantity with totalized cut-off
Pressure Monitoring	Discharge PT / local gauge
Injection Control	Fine control valve or check/injection valve as required
Remote Control	PLC/DCS/SCADA command and feedback

12. LOW-FLOW ACCURACY

At very low flow rates, system accuracy depends on pump displacement, motor speed resolution, meter rangeability, temperature/viscosity stability and the absence of gas bubbles. The skid shall therefore be calibrated over the actual operating range rather than at a single nominal condition.

MATERIALS, SEALS & CHEMICAL COMPATIBILITY

13. MATERIAL SELECTION

Duty	Typical Material Options
Hydrocarbon / Fuel Additives	SS316L, alloy steel or compatible stainless construction
Solvents / Specialty Chemicals	SS316L or compatible alloy, selected elastomers
Lubricant Additives	SS316L or suitable alloy; heated construction optional
Corrosive Specialty Chemicals	Special alloys or corrosion-resistant materials by compatibility review

14. SEALING OPTIONS

Mechanical-seal, lip-seal or magnetically coupled/sealless designs may be used depending on pump manufacturer and service. Magnetic-drive construction is preferred where zero visible shaft leakage is a key requirement and the fluid is compatible with the containment shell and internal bearings.

15. CHEMICAL COMPATIBILITY

Pump gears, bearings, shaft, body, seals, gaskets, tubing and flowmeter wetted parts shall all be checked against the process liquid concentration and temperature. Abrasive solids, crystallizing chemicals and polymerizing fluids require special review and may be better served by another LINESIG dosing family.

AUTOMATION, SAFETY & ELECTRICAL

16. PLC / HMI FUNCTIONS

- Manual / automatic / remote operating modes
- Pump speed setpoint and feedback
- Flow setpoint, actual flow and totalization
- Ratio or batch dosing logic
- Low tank level inhibit
- High discharge pressure trip
- Low / no-flow alarm
- Filter differential pressure alarm where provided
- Motor / VFD fault monitoring
- Remote ESD and process permissive interface

17. SAFETY FEATURES

Protection	Typical Function
Pressure Relief / Bypass	Protects positive-displacement pump and discharge system
High Pressure Trip	Stops pump on abnormal backpressure
Low Suction Level / Pressure	Prevents dry operation or cavitation
Fine Suction Filter	Protects close-clearance pumping elements
Leak / Drip Detection	Optional for hazardous or environmentally sensitive duty
Emergency Stop	Local and optional remote ESD
Earthing / Bonding	Applied to flammable service where required

18. HAZARDOUS AREA

For hydrocarbon, solvent or flammable additive service, motors, VFD arrangements, junction boxes, transmitters, flowmeters and control devices shall be selected to suit the project hazardous-area classification. ATEX/IECEx certification requirements are project-specific.

APPLICATION CONFIGURATIONS & FAMILY SELECTION

19. TYPICAL CONFIGURATIONS

Configuration	Recommended Use
1 x 100%	Single precision dosing point
1W + 1S	Critical continuous low-flow injection
2W + 1S	Two active streams with common standby philosophy
Multi-Line Modular Skid	Independent additives or injection points
Day Tank Package	Complete local storage + precision dosing
Heated Skid	Viscous additives requiring temperature conditioning

20. WHEN TO SELECT LNS-DZ-PDG

- Very low-flow, high-accuracy dosing is required.
- A smooth, low-pulsation flow is preferred.
- The liquid is clean and sufficiently lubricating for precision gear elements.
- Fine additive, catalyst, dye, marker or specialty-chemical injection is required.
- Wide electronic turndown through speed control is advantageous.
- Compact multi-line dosing is required.

21. WHEN ANOTHER FAMILY MAY BE BETTER

Duty Requirement	Recommended Family
Medium/high-flow smooth dosing or blending	LNS-DZ-CEN - Centrifugal
Corrosive / leak-sensitive chemical metering	LNS-DZ-PDD - Diaphragm
High-pressure injection	LNS-DZ-PDP - Piston / Plunger
Very low-flow precision clean-liquid dosing	LNS-DZ-PDG - Precision Micro Gear

ORDERING DATA & STANDARD SCOPE

22. MINIMUM DATA REQUIRED FOR FINAL SIZING

- Chemical / additive name and concentration
- Minimum / normal / maximum required flow
- Normal and maximum discharge pressure
- Suction source, available pressure and liquid level range
- Density and viscosity at minimum / normal / maximum temperature
- Vapor pressure and presence of dissolved or entrained gas
- Maximum allowable particle size / cleanliness requirement
- Required accuracy, repeatability and turndown
- Number of dosing lines and standby philosophy
- Required wetted material and seal arrangement
- Hazardous-area classification
- Control mode: local, ratio, batch, PLC/DCS or remote
- Power supply, communication protocol and applicable project standards

23. TYPICAL STANDARD SCOPE

Item	Typical Supply
A	Skid structural frame and supports
B	Precision micro gear pump(s) and motor/servo drive
C	Suction isolation and fine filtration
D	Common or individual suction manifold
E	Pressure relief / bypass arrangement
F	Flowmeter / flow verification system
G	Discharge pressure instrumentation
H	Fine control / injection / check valves as required
I	Day tank or small buffer vessel when specified
J	Local PLC/HMI or junction box when specified
K	Internal piping, tubing and cable supports
L	Documentation: datasheet, GA, P&ID, I/O list, control philosophy and test records as applicable



PRECISION FLOW. SMOOTH INJECTION. REPEATABLE PERFORMANCE.

LINE SIG LNS-DZ-PDG — engineered for accurate low-flow additive and specialty-chemical dosing.